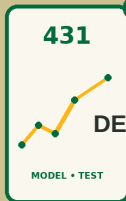
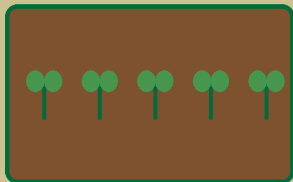


Calculate Design Flow Rate



CALCULATE DESIGN FLOW RATE

DESIGN • INSTRUMENT • MODEL • EVALUATE

DESIGN → INSTRUMENT → MODEL → EVALUATE → OPTIMIZE

TECHNICAL INVESTIGATION

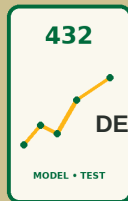
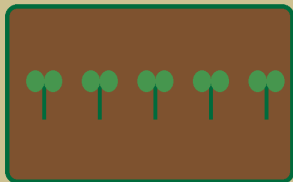
Use the specified measurements, equation, and assumptions. Show units throughout, calculate the result, and interpret its engineering or production meaning.

EVALUATE & DECIDE

State the result, uncertainty or limitation, and the next design, measurement, or management decision supported by the evidence.

DEFINE → MEASURE → MODEL → TEST → DECIDE → IMPROVE

Calculate Static Head



CALCULATE STATIC HEAD

DESIGN • INSTRUMENT • MODEL • EVALUATE

DESIGN → INSTRUMENT → MODEL → EVALUATE → OPTIMIZE

TECHNICAL INVESTIGATION

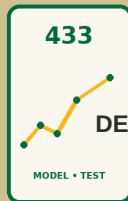
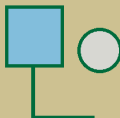
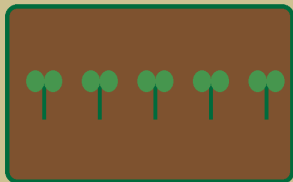
Use the specified measurements, equation, and assumptions. Show units throughout, calculate the result, and interpret its engineering or production meaning.

EVALUATE & DECIDE

State the result, uncertainty or limitation, and the next design, measurement, or management decision supported by the evidence.

DEFINE → MEASURE → MODEL → TEST → DECIDE → IMPROVE

Estimate Friction Loss



ESTIMATE FRICTION LOSS

DESIGN • INSTRUMENT • MODEL • EVALUATE

DESIGN → INSTRUMENT → MODEL → EVALUATE → OPTIMIZE

TECHNICAL INVESTIGATION

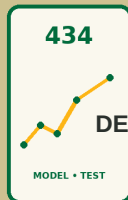
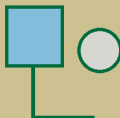
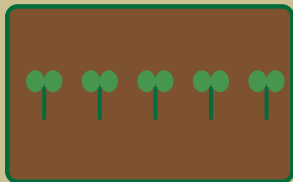
Use the specified measurements, equation, and assumptions. Show units throughout, calculate the result, and interpret its engineering or production meaning.

EVALUATE & DECIDE

State the result, uncertainty or limitation, and the next design, measurement, or management decision supported by the evidence.

DEFINE → MEASURE → MODEL → TEST → DECIDE → IMPROVE

Compare Pipe Diameters



COMPARE PIPE DIAMETERS

DESIGN • INSTRUMENT • MODEL • EVALUATE

DESIGN → INSTRUMENT → MODEL → EVALUATE → OPTIMIZE

TECHNICAL INVESTIGATION

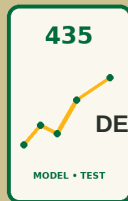
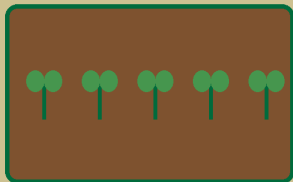
Use quantitative evidence and stated criteria. Compare alternatives or conditions, account for uncertainty, and justify the conclusion.

EVALUATE & DECIDE

State the result, uncertainty or limitation, and the next design, measurement, or management decision supported by the evidence.

DEFINE → MEASURE → MODEL → TEST → DECIDE → IMPROVE

Measure Emitter Discharge



MEASURE EMITTER DISCHARGE

DESIGN • INSTRUMENT • MODEL • EVALUATE

DESIGN → INSTRUMENT → MODEL → EVALUATE → OPTIMIZE

TECHNICAL INVESTIGATION

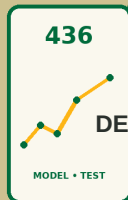
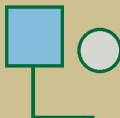
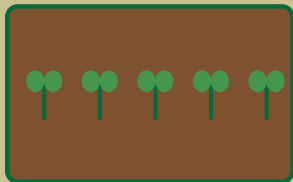
Complete the investigation using explicit assumptions, systematic measurements, and reproducible records. Interpret the evidence and state the decision implication.

EVALUATE & DECIDE

State the result, uncertainty or limitation, and the next design, measurement, or management decision supported by the evidence.

DEFINE → MEASURE → MODEL → TEST → DECIDE → IMPROVE

Calculate Distribution Uniformity



CALCULATE DISTRIBUTION UNIFORMITY

DESIGN • INSTRUMENT • MODEL • EVALUATE

DESIGN → INSTRUMENT → MODEL → EVALUATE → OPTIMIZE

TECHNICAL INVESTIGATION

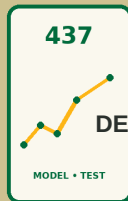
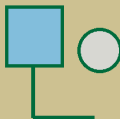
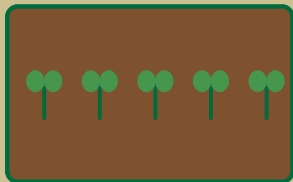
Use the specified measurements, equation, and assumptions. Show units throughout, calculate the result, and interpret its engineering or production meaning.

EVALUATE & DECIDE

State the result, uncertainty or limitation, and the next design, measurement, or management decision supported by the evidence.

DEFINE → MEASURE → MODEL → TEST → DECIDE → IMPROVE

Calculate Irrigation Set Time



CALCULATE IRRIGATION SET TIME

DESIGN • INSTRUMENT • MODEL • EVALUATE

DESIGN → INSTRUMENT → MODEL → EVALUATE → OPTIMIZE

TECHNICAL INVESTIGATION

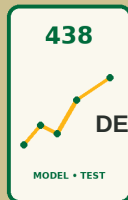
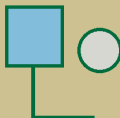
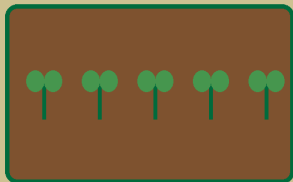
Use the specified measurements, equation, and assumptions. Show units throughout, calculate the result, and interpret its engineering or production meaning.

EVALUATE & DECIDE

State the result, uncertainty or limitation, and the next design, measurement, or management decision supported by the evidence.

DEFINE → MEASURE → MODEL → TEST → DECIDE → IMPROVE

Estimate Pump Power



ESTIMATE PUMP POWER

DESIGN • INSTRUMENT • MODEL • EVALUATE

DESIGN → INSTRUMENT → MODEL → EVALUATE → OPTIMIZE

TECHNICAL INVESTIGATION

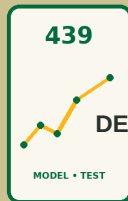
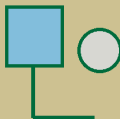
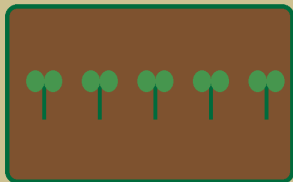
Use the specified measurements, equation, and assumptions. Show units throughout, calculate the result, and interpret its engineering or production meaning.

EVALUATE & DECIDE

State the result, uncertainty or limitation, and the next design, measurement, or management decision supported by the evidence.

DEFINE → MEASURE → MODEL → TEST → DECIDE → IMPROVE

Calculate Pump Efficiency



CALCULATE PUMP EFFICIENCY

DESIGN • INSTRUMENT • MODEL • EVALUATE

DESIGN → INSTRUMENT → MODEL → EVALUATE → OPTIMIZE

TECHNICAL INVESTIGATION

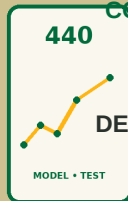
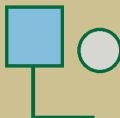
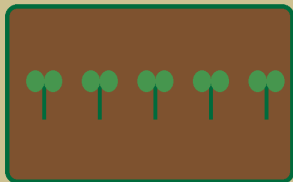
Use the specified measurements, equation, and assumptions. Show units throughout, calculate the result, and interpret its engineering or production meaning.

EVALUATE & DECIDE

State the result, uncertainty or limitation, and the next design, measurement, or management decision supported by the evidence.

DEFINE → MEASURE → MODEL → TEST → DECIDE → IMPROVE

Complete an Irrigation Design Review



COMPLETE AN IRRIGATION DESIGN REVIEW

DESIGN • INSTRUMENT • MODEL • EVALUATE

DESIGN → INSTRUMENT → MODEL → EVALUATE → OPTIMIZE

TECHNICAL INVESTIGATION

Define the objective, boundary, variables, constraints, and performance criteria. Develop a testable design or model and document the assumptions.

EVALUATE & DECIDE

State the result, uncertainty or limitation, and the next design, measurement, or management decision supported by the evidence.

DEFINE → MEASURE → MODEL → TEST → DECIDE → IMPROVE